

Demonstration of Proposed Features

Date	06 July 2026
Team ID	
Project Name	Voice Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Speech-to-Text Transcription	Converts the uploaded/recorded audio explanation into text using OpenAI Whisper.	Implemented	Yes	Tested with .wav, .mp3 and .m4a inputs.
2	Semantic Similarity Scoring	Compares the transcript against the reference concept definition using Sentence-BERT embeddings.	Implemented	Yes	Includes keyword gap analysis (covered vs. missing terms).
3	Audio Fluency Analysis	Extracts pause ratio, filler word count, pitch and RMS energy from the audio using Librosa.	Implemented	Yes	Covers 18 common filler words/phrases.
4	Weighted Scoring & Feedback	Combines semantic (60%) and fluency (40%) scores into a final score with auto-generated written feedback.	Implemented	Yes	Labels responses as Strong / Moderate / Poor Understanding.
5	PDF Report Generation	Generates a downloadable PDF report with scores, transcript, metrics and charts using ReportLab.	Implemented	Yes	Includes waveform and mel-spectrogram visuals.
6	Interactive Web Dashboard	Web UI for uploading audio and viewing results via waveform, radar and donut charts.	Implemented	Yes	Supports drag-and-drop upload and dark/light theme.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%